

Q5 Smart City IoT Energy Management System
10 marks

Your Code

Input

3
101
Arun
85 90 80

Output

```

26     calculateAverage();
27     determinePerformance();
28     totalStudents++;
29 }
30
31 void calculateAverage() {
32     float sum = 0;
33
34     for (int i = 0; i < 3; i++) {
35         sum += marks[i];
36     }
37
38     average = sum / 3.0;
39 }
40
41 void determinePerformance() {
42     if (average >= 80)
43         performanceLevel = "Excellent";
44     else if (average >= 60)
45         performanceLevel = "Good";
46     else if (average >= 40)
47         performanceLevel = "Average";
48     else
49         performanceLevel = "Needs Improvement";
50 }
51
52 void display() {
53     cout << "Student ID: " << studentID << endl;
54     cout << "Name: " << name << endl;
55     cout << "Average: " << fixed << setprecision(2) << average << endl;
56     cout << "Performance Level: " << performanceLevel << endl;
57     cout << "-----" << endl;
58 }
59
60 static void displayTotalStudents() {
61     cout << "Total Students Analyzed: "
62         << totalStudents << endl;
63 }
64
65
66 Student::totalStudents = 0;
67
68 main() {

```

```

53     cout << "Student ID: " << studentID << endl;
54     cout << "Name: " << name << endl;
55     cout << "Average: " << fixed << setprecision(2) << average << endl;
56     cout << "Performance Level: " << performanceLevel << endl;
57     cout << "-----" << endl;
58 }
59
60 static void displayTotalStudents() {
61     cout << "Total Students Analyzed: "
62         << totalStudents << endl;
63 }
64
65
66 Student::totalStudents = 0;
67
68 main() {
69     int n;
70     cin >> n;
71
72     Student students[100];
73
74     for (int i = 0; i < n; i++) {
75         students[i].input();
76     }
77
78     cout << "\nSTUDENT PERFORMANCE REPORT\n";
79     cout << "-----\n";
80
81     for (int i = 0; i < n; i++) {
82         students[i].display();
83     }
84
85     Student::displayTotalStudents();
86
87     return 0;
88 }

```

Digital Healthcare Patient Monitoring System

10 marks

Digital Healthcare Patient Monitoring System Scenario: Smart Healthcare. A modern hospital wants to implement a Digital Patient Monitoring System that maintains patient information and automatically calculates basic health-risk indicators from recorded medical parameters.

- A. Design a class Patient with attributes such as Patient ID, Name, Age, Heart Rate, Blood Pressure, and Health Status.
- B. Implement member functions to input patient details, calculate/identify the health status based on predefined conditions, and display the patient report.
- C. Store multiple patient records using an array of objects.
- D. Use a static member to track the total number of patients monitored.
- E. Demonstrate data hiding, encapsulation, and structured object-oriented program design.

Your Code

```
1 #include <iostream>
2 using namespace std;
3
4 class Patient {
5     int id, age, heartRate, bp;
6     string name, status;
7
8 public:
9     static int total;
10
11     void input() {
12         cin >> id >> name >> age >> heartRate >> bp;
13         check();
14         total++;
15     }
16
17     void check() {
18         if (heartRate >= 60 && heartRate <= 100 && bp >= 80 && bp <= 120)
19             status = "Normal";
20         else
21             status = "At Risk";
22     }
23
24     void display() {
25         cout << "ID: " << id << endl;
26         cout << "Name: " << name << endl;
```

Input

```
2
101 arun 25 72 110
102 priya 45 115 140
```

Output



1/1 answered

```
16
17 void check() {
18     if (heartRate >= 60 && heartRate <= 100 && bp >= 90 && bp <= 110)
19         status = "Normal";
20     else
21         status = "At Risk";
22 }
23
24 void display() {
25     cout << "ID: " << id << endl;
26     cout << "Name: " << name << endl;
27     cout << "Age: " << age << endl;
28     cout << "Heart Rate: " << heartRate << endl;
29     cout << "Blood Pressure: " << bp << endl;
30     cout << "Health Status: " << status << endl;
31 }
32 };
33
34 int Patient::total = 0;
35
36 int main() {
37     int n;
38     cin >> n;
39
40     Patient p[50];
41
42     for(int i=0; i<n; i++)
43         p[i].input();
44
45     for(int i=0; i<n; i++)
46         p[i].display();
47
48     cout << "Total Patients: " << Patient::total;
49
50     return 0;
51 }
```


Q5 Smart City IoT Energy Management System 10 marks

Input

2
101 Arun 9876543210 5000
102 Priya9876543211 3000
1000 500

Output

```
19 }
20
21 void withdraw(float x) {
22     if (x <= balance) balance -= x;
23     else cout << "Insufficient Balance\n";
24 }
25
26 void display() {
27     cout << "Wallet ID: " << id << endl;
28     cout << "Name: " << name << endl;
29     cout << "Mobile: " << mobile << endl;
30     cout << "Balance: " << balance << endl;
31 }
32 };
33
34 int DigitalWallet::total = 0;
35
36 int main() {
37     int n;
38     cin >> n;
39
40     DigitalWallet w[50];
41
42     for(int i=0;i<n;i++)
43         w[i].create();
44
45     for(int i=0;i<n;i++) {
46         float d, x;
47         cin >> d >> x;
48         w[i].deposit(d);
49         w[i].withdraw(x);
50         w[i].display();
51     }
52
53     cout << "Active Wallets: " << DigitalWallet::total;
54     return 0;
55 }
```

QUESTIONS

Q1
AI-Based Student Performance
Analytics System
10 marks

Q2
Digital Healthcare Patient
Monitoring System
10 marks

Q3
Digital Wallet and FinTech
Transaction System
10 marks

Q4
Smart Agriculture Crop
Monitoring System
10 marks

Q5
Smart City IoT Energy
Management System
10 marks

5/5 answered

Q4 Smart Agriculture Crop Monitoring System

10 marks

Smart Agriculture Crop Monitoring System Scenario: IoT-Based Agriculture. An agricultural research centre is developing an IoT-based Smart Agriculture System to monitor crop fields using sensor data and identify whether crops require irrigation.

- A. Design a class CropField with attributes such as Field ID, Crop Name, Soil Moisture, Temperature, Humidity, and Irrigation Status.
- B. Implement member functions to input sensor readings, evaluate soil conditions, determine irrigation requirements, and display field information.
- C. Use an array of objects to maintain information about multiple crop fields.
- D. Include a static member to count the total number of monitored fields.
- E. Ensure proper encapsulation and modular object-oriented design suitable for an IoT-based agricultural application.

Your Code

```
1 #include <iostream>
2 using namespace std;
3
4 class CropField {
5     int id;
6     string crop, status;
7     float moisture, temp, humidity;
8
9 public:
10     static int total;
11
12     void input() {
13         cin >> id >> crop >> moisture >> temp >> humidity;
14         check();
15         total++;
16     }
17
18     void check() {
19         if (moisture < 40)
20             status = "Irrigation Required";
21         else
22             status = "No Irrigation";
23     }
24
25     void display() {
26         cout << "Field ID: " << id << endl;
27         cout << "Crop: " << crop << endl;
```

Input

```
2
101 Rice 35 30 70
102 wheat 55 28 65
```

Output


```
16 }
17
18 void check() {
19     if (moisture < 40)
20         status = "Irrigation Required";
21     else
22         status = "No Irrigation";
23 }
24
25 void display() {
26     cout << "Field ID: " << id << endl;
27     cout << "Crop: " << crop << endl;
28     cout << "Soil Moisture: " << moisture << endl;
29     cout << "Temperature: " << temp << endl;
30     cout << "Humidity: " << humidity << endl;
31     cout << "Irrigation Status: " << status << endl;
32 }
33 };
34
35 int CropField::total = 0;
36
37 int main() {
38     int n;
39     cin >> n;
40
41     CropField f[50];
42
43     for(int i=0;i<n;i++)
44         f[i].input();
45
46     for(int i=0;i<n;i++)
47         f[i].display();
48
49     cout << "Total Fields: " << CropField::total;
50
51     return 0;
52 }
```

QUESTIONS

- Q1 AI-Based Student Performance Analytics System 10 marks
- Q2 Digital Healthcare Patient Monitoring System 10 marks
- Q3 Digital Wallet and FinTech Transaction System 10 marks
- Q4 Smart Agriculture Crop Monitoring System 10 marks
- Q5 Smart City IoT Energy Management System 10 marks

5/5 answered

Q5 Smart City IoT Energy Management System

10 marks

Smart City IoT Energy Management System Scenario: Sustainable Smart City. A smart-city research project aims to develop an IoT-based Energy Management System for monitoring electricity consumption in different buildings and identifying buildings with high energy usage.

- A. Create a class Building with attributes such as Building ID, Building Name, Building Type, Energy Consumption, and Energy Status.
- B. Implement member functions to input building details, calculate energy status based on consumption, and display energy usage information.
- C. Use an array of objects to manage multiple buildings in the smart-city system.
- D. Include a static member to maintain the total number of buildings monitored.
- E. Apply encapsulation, data hiding, and structured C++ object-oriented programming principles in the implementation.

Your Code

```
1 #include <iostream>
2 using namespace std;
3
4 class Building {
5 private:
6     int buildingID;
7     string buildingName, buildingType, energyStatus;
8     float energyConsumption;
9
10 public:
11     static int totalBuildings;
12
13     void input() {
14         cin >> buildingID >> buildingName >> buildingType
15         >> energyConsumption;
16
17         calculateStatus();
18         totalBuildings++;
19     }
20
21     void calculateStatus() {
22         if (energyConsumption <= 500)
23             energyStatus = "Low";
24         else if (energyConsumption <= 1000)
25             energyStatus = "Medium";
26         else
27             energyStatus = "High";
```

Input

```
3
101 collage education 450
102 hospital medical 850
103 mall commercial 1200
```

Output



```
27         energyStatus = "High";
28     }
29
30     void display() {
31         cout << "Building ID: " << buildingID << endl;
32         cout << "Building Name: " << buildingName << endl;
33         cout << "Building Type: " << buildingType << endl;
34         cout << "Energy Consumption: " << energyConsumption << endl;
35         cout << "Energy Status: " << energyStatus << endl;
36         cout << "-----" << endl;
37     }
38 };
39
40 int Building::totalBuildings = 0;
41
42 int main() {
43     int n;
44     cin >> n;
45
46     Building b[50];
47
48     for (int i = 0; i < n; i++) {
49         b[i].input();
50     }
51
52     cout << "SMART CITY ENERGY REPORT" << endl;
53     cout << "===== " << endl;
54
55     for (int i = 0; i < n; i++) {
56         b[i].display();
57     }
58
59     cout << "Total Buildings Monitored: "
60         << Building::totalBuildings << endl;
61
62     return 0;
63 }
```